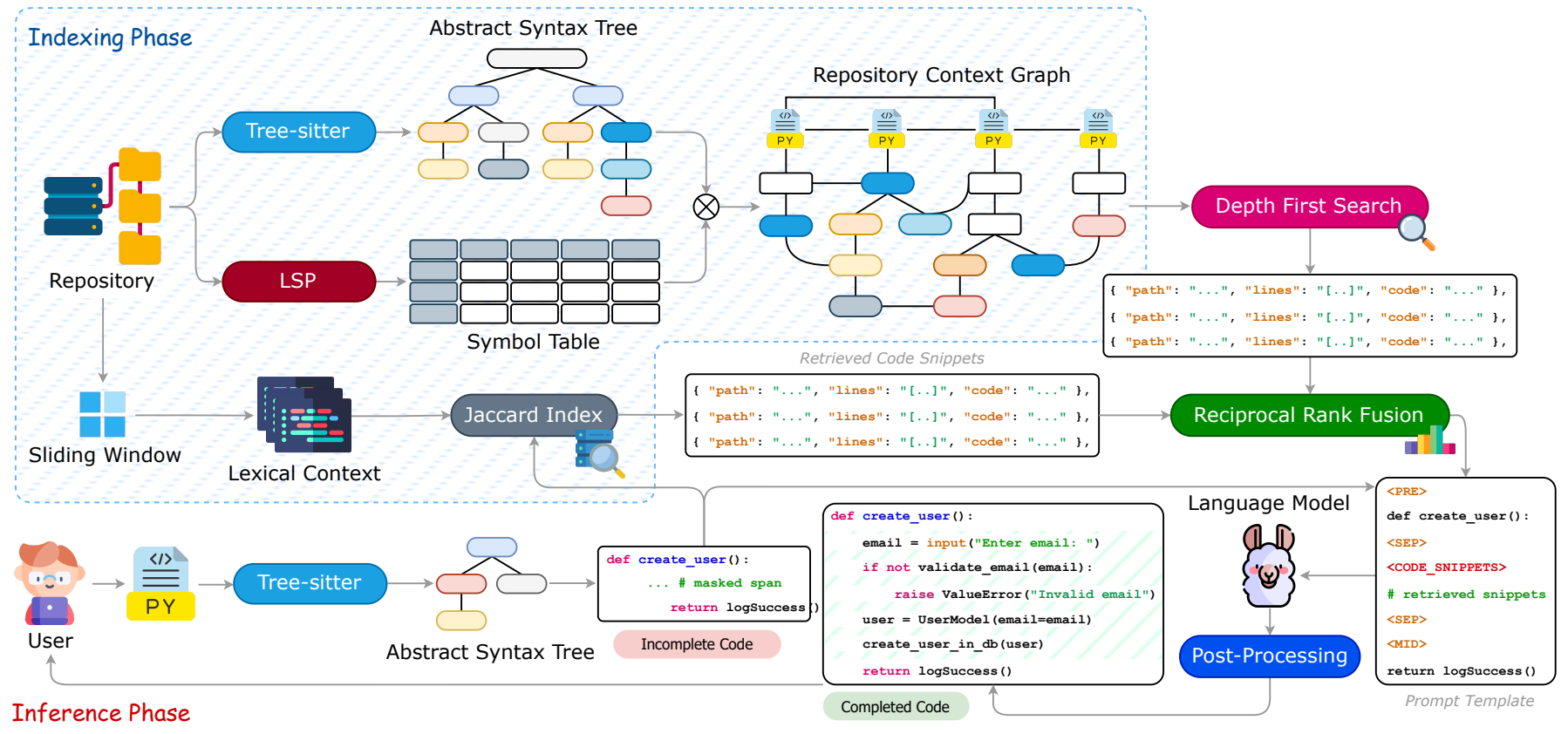

Indexing Phase
Abstract Syntax Tree
Repository Context Graph
Tree-sitter
Repository
LSP
Symbol Table
Depth First Search
{ "path": "...", "lines": "[..]", "code": "..." },
{ "path": "...", "lines": "[..]", "code": "..." },
{ "path": "...", "lines": "[..]", "code": "..." },
Retrieved Code Snippets
Sliding Window
Lexical Context
Jaccard Index
Reciprocal Rank Fusion
Language Model
<PRE>
def create_user():
<SEP>
<CODE_SNIPPETS>
retrieved snippets
<SEP>
<MID>
return logSuccess()
Prompt Template
User
Tree-sitter
Abstract Syntax Tree
def create_user():
... # masked span
return logSuccess()
Incomplete Code
def create_user():
email = input("Enter email: ")
if not validate_email(email):
raise ValueError("Invalid email")
user = UserModel(email=email)
create_user_in_db(user)
return logSuccess()
Completed Code
Post-Processing
Inference Phase